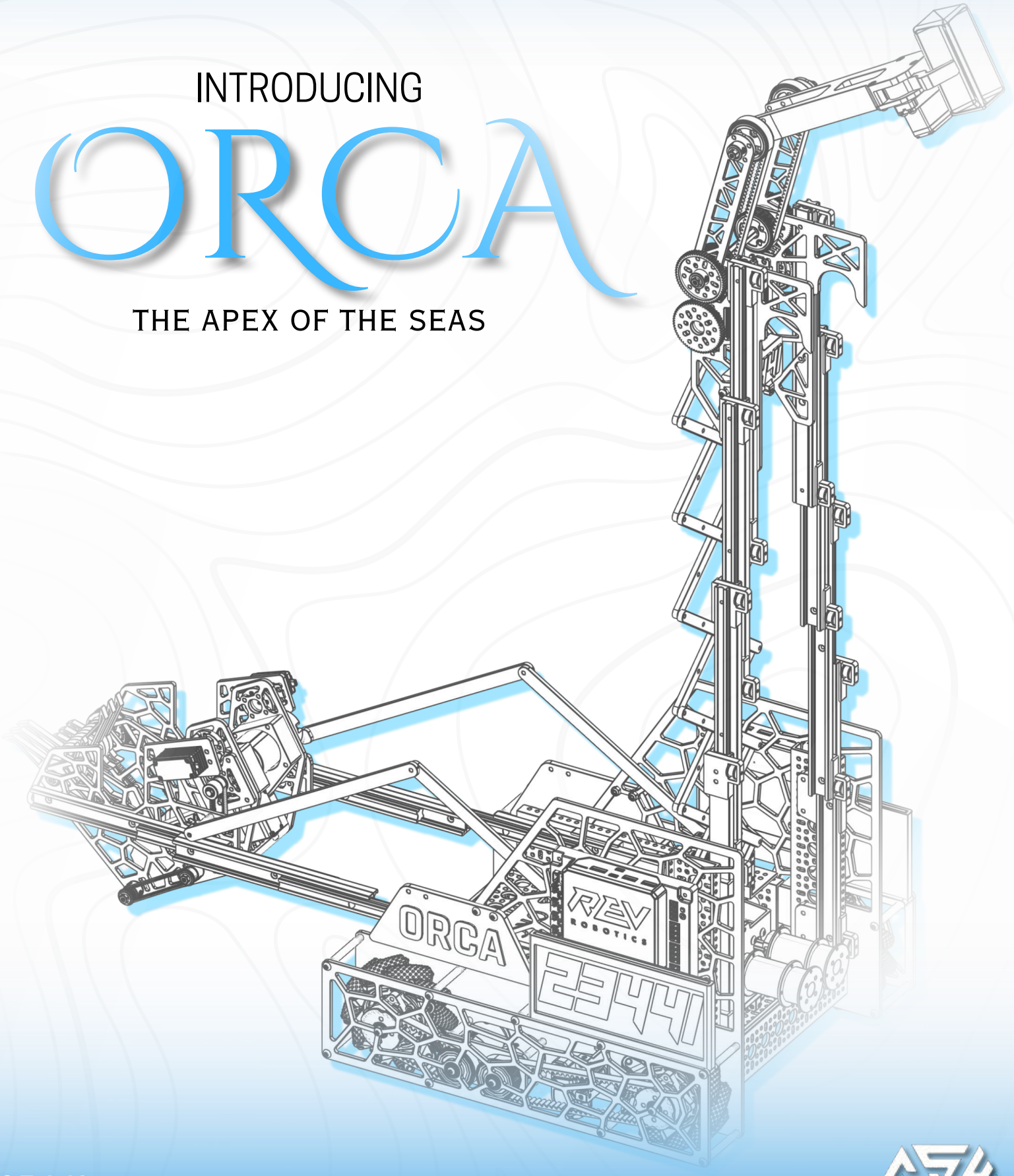
INTRODUCING

ORCA

THE APEX OF THE SEAS



23441

Team Intro

Team Overview

We are **Team #23441**, a student-led FTC Robotics team based in Cerritos, California. Our team is **entirely run by students** focusing on innovation, teamwork, and STEM outreach.

We are dedicated to building **competitive** robots and fostering a love for engineering and programming **along our journey.**





Suryaa R.
Design Captain



Pranav A.
Program Captain



Sachin S.
Mechanics Captain



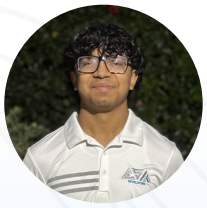
Krish B.
Program Member



Ankit C.
Mechanics Member



Aahan N.
Program Member



Anish R.
Mechanics Member

Our Coach



Mr. Chandra
Guides the team in strategy, leadership, and preparation.

Mission Statement

To create a **competitive** robotics team that excels in design and programming while promoting **STEM education** and **teamwork** in the community.

Team Goals



My goal is to lead our robotics team with **dedication**, fostering teamwork and new ideas through **collaboration**, hard work, and a shared passion for **innovation**.

Suryaa R.

My goal is to help our robotics team push the **boundaries** of innovation while **mentoring** new members to ensure they feel supported, engaged, and **empowered** to contribute.

Pranav A.





My goal is to shine with **curiosity**, solving problems with creativity, while **learning** and growing through each **opportunity**.

Sachin S.

My goal is to rise with **determination**, fueling progress through dedication, while **mastering** challenges with patience and sparking new **creations**.

Krish B.





My goal is to grow with **ambition**, while mastering skills with **persistence** and fueling **our team's** innovation.

Ankit C.

My goal is to excel with **passion**, sparking progress with each action, while **building** strength through focus and **dedication** to our team's **satisfaction**.

Aahan N.





My goal is to fit my **vocation**, inspiring growth and **precision** through **effort** and **vision**.

Anish R.

Structure/Flow of Information



Coach Chandra

Coach:

Coach **Chandra** provides the team with **guidance**, leadership, and **mentorship** to the team. He helps organize meetings, facilitate collaboration, and ensure we follow competition guidelines and **deadlines**. His role is to **support** the team in developing skills and **solving** challenges.



Suryaa R.



Pranav A.



Sachin S.

Captains:

Our captains **oversee** different aspects of the team, like engineering, programming, and outreach. They help **coordinate** tasks, manage **communication** between sub-teams, and make sure everyone stays on track. Their job is to **lead by example** and keep the team working smoothly.



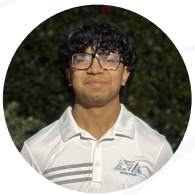
Ankit C.



Aahan N.



Krish B.



Anish R.

Members:

Our members **contribute** to various roles like designing, building, programming, and **documenting** the robot. They also help with outreach, fundraising, and presentations. Everyone has a **part to play** in making the team **successful**.

Industry Mentors



Lingareddy N.

Senior Engineering Leader at **Walmart Global Tech**



Murugan A.

Senior Navigation Development Engineer at **Mercedes**

Our **industry** mentors played a crucial role in our team's growth by providing **technical** expertise, strategic **guidance**, and real-world engineering insights.

Goutham N.

Solutions Senior Leader at **Amazon Web Services**



Kiran C.

Sr. Director of Engineering at **Elevance Health**



Orca Overview

Outtake Claw Module

Coaxial Virtual Four Bar

Ascent Hooks

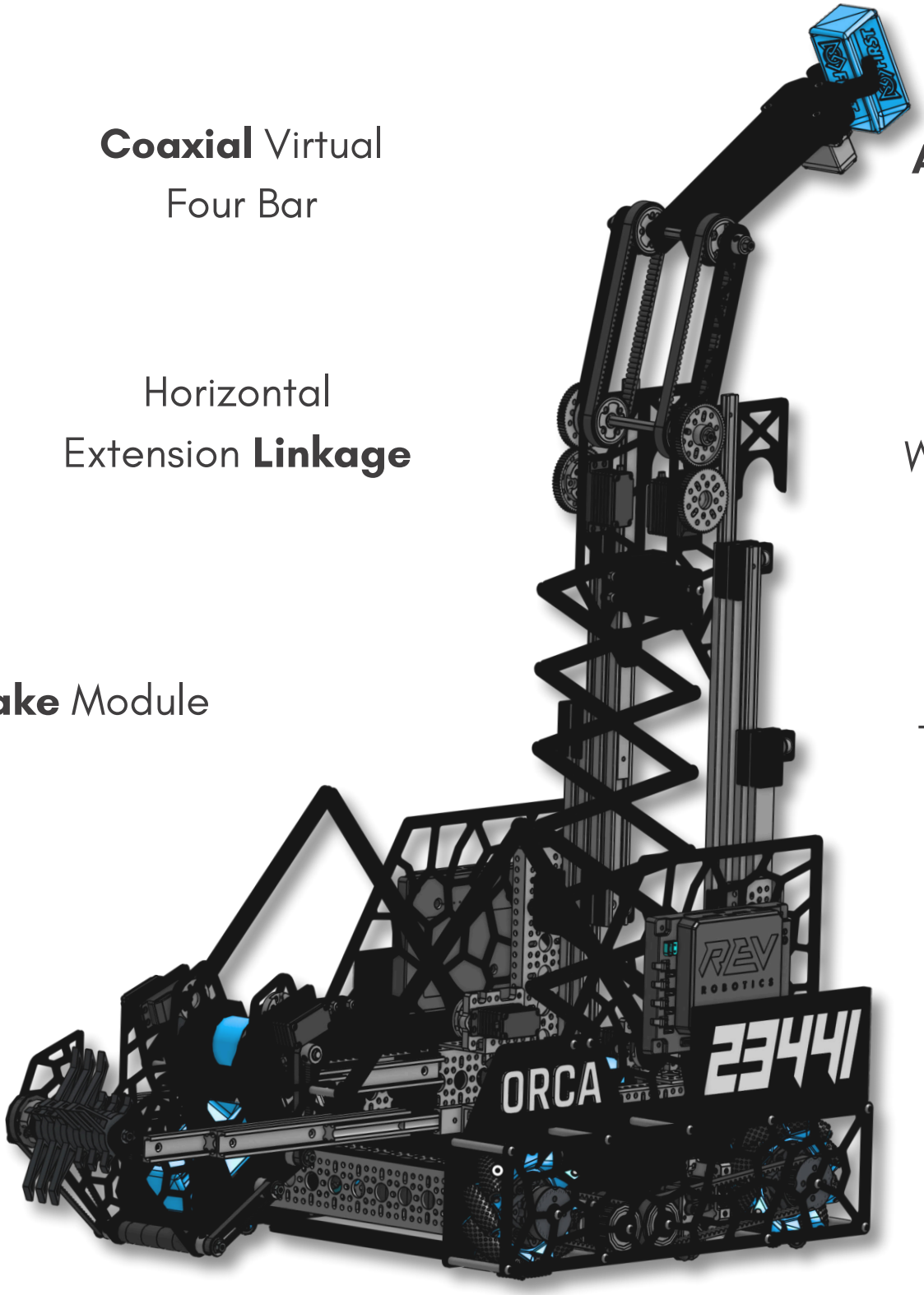
Horizontal Extension **Linkage**

Wiring **Scissor Linkage**

Intake Module

Team **Number Plate**

Butterfly Drivetrain



Game Strategy

Samples:

After **extensive** research and testing, we determined that each cycle of collecting and depositing a sample into the high basket takes **approximately** 9 seconds. Based on these calculations, we estimate that we can score up to 13 samples during the tele-op period, **yielding** a total of 104 points. By refining our intake and outtake **mechanisms**, we aim to improve consistency, reduce cycle time, and maximize our scoring **efficiency** throughout the **match**.

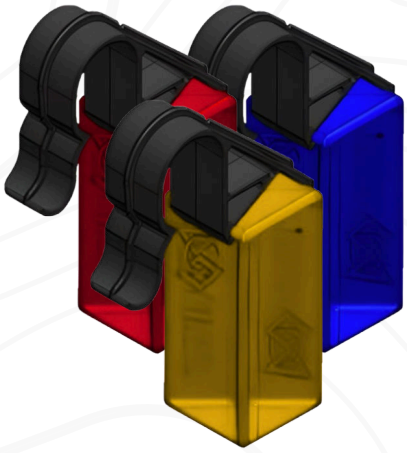
Specimen:

Through **additional** calculations, we also determined that the process of collecting a **sample**, delivering it to the **human** player, converting it into a specimen, and depositing it onto the high rung takes **approximately** 15 seconds. This **estimation** allows us to score around eight specimens during the tele-op period, **resulting** in a total of 80 points. By optimizing efficiency and minimizing cycle **times**, we aim to maximize our scoring **potential** throughout the match.



Overall Strategy:

Based on our **analysis**, we prioritized sample-based scoring for its efficiency and higher point **potential**. However, if our alliance partners excel at sample scoring, we can **switch** to specimen-based scoring as needed. In autonomous, we focus on specimens since the **observation** zone is closer to the submersible than the basket, ensuring a more consistent **strategy**. This approach maximizes adaptability and scoring potential while helping us **contribute** to victory.



Autonomous:

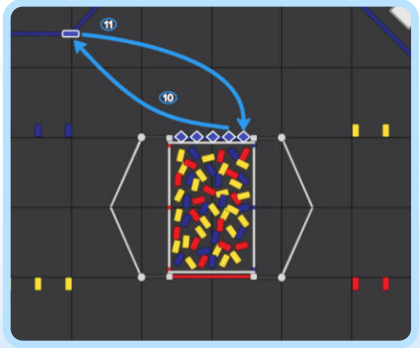
For the **autonomous** period, we developed two programs: one for specimen-based **scoring** and one for sample-based scoring. The specimen-based autonomous **follows** a 5+0 strategy, prioritizing specimen **placement**, while the sample-based autonomous follows a 0+4 **strategy**, focusing **solely** on sample scoring.

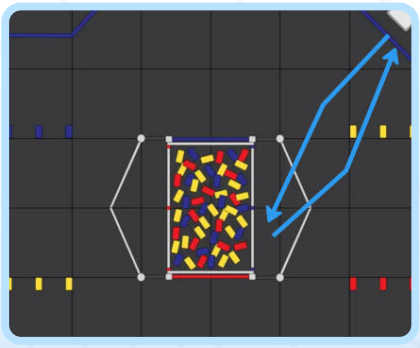
Tele-op:

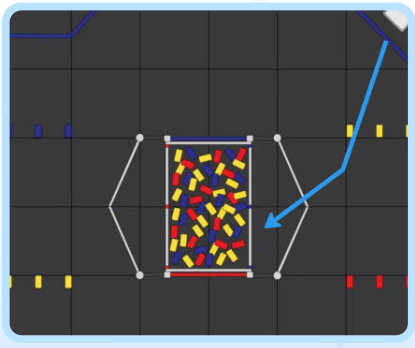
Since we have **committed** to sample-based scoring during the tele-op period, our slides allow **precise** claw positioning over the basket for controlled sample **deposition**. This ensures **efficiency** throughout the match and by refining our movements and **minimizing** errors, we optimize cycle times and maximize scoring **potential**.

Endgame:

During the **endgame**, our strategy focuses on efficiently scoring two samples before **executing** a level two ascent to maximize points. By ensuring a smooth **transition** between **scoring** and climbing, we optimize our endgame performance while **maintaining** stability and reliability in our final actions.







Design Process

Define
It is important we define the **objective** of the game and design our robot **accordingly**.

Ideate
Brainstorming and **generating** ideas in order to reach our **goal**.

Prototype
Sketching and building a sample **prototype** to put our ideas to the **test**.

Build
Actually build the design **based** on the prototype and **new** ideas.

Analyze
Analyze our built product and see if other **iterations** are needed to progress **further**.

Sketching:

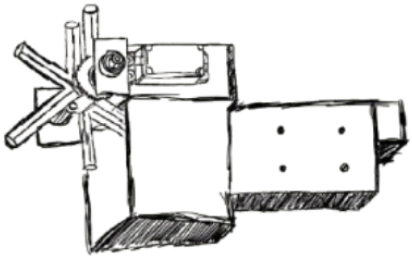
To begin our design **journey**, the design process often begins with sketches that help visualize ideas. These **initial** drawings allow us to explore different concepts before committing to a **final** design.

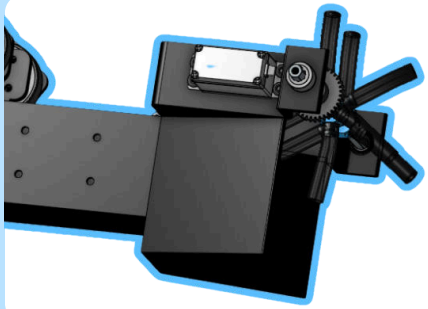
Cadding:

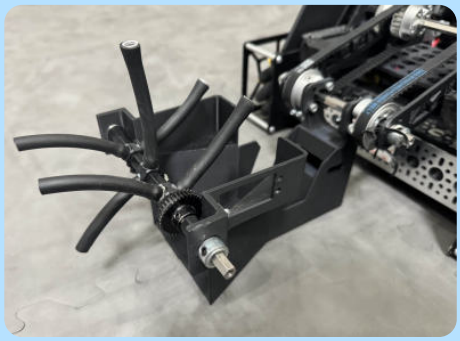
Computer-Aided Design (CAD) is **essential** for creating precise digital models. This step ensures that parts fit together **correctly** and **function** as intended before manufacturing.

Building:

Building the ORCA involved **assembling** components based on the CAD model. We used a **variety** of materials, from metal and plastic to advanced composites, ensuring **durability** and performance.







Intake:

Our intake uses an **active** system to pull in samples quickly, reducing the need for **precise** driver control. This allows for faster and more **consistent** collection, keeping the robot efficient during matches. The design **prioritizes** speed and reliability, ensuring smooth operation without jams or delays. With a quick and **controlled** intake, we **maximize** performance and maintain a competitive edge.

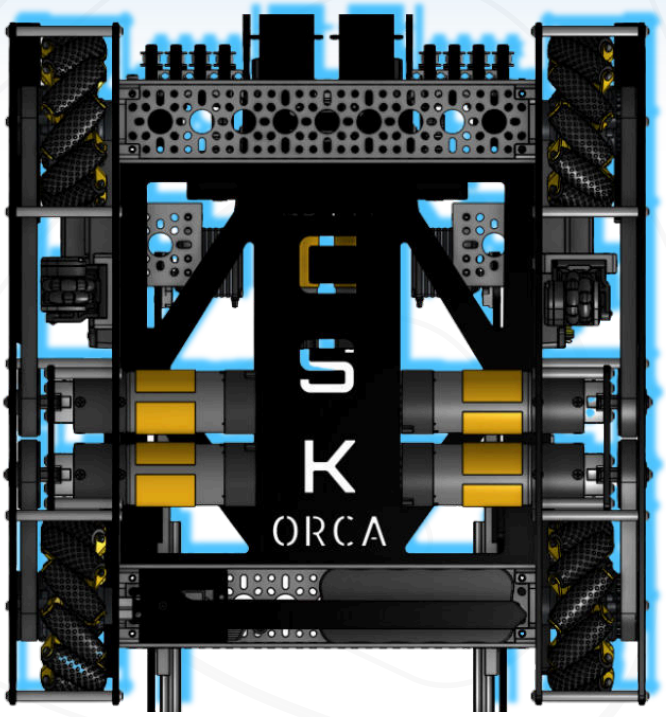
Outtake:

Our outtake **seamlessly** collects samples from the intake, ensuring a smooth transfer for scoring. With two **degrees** of freedom, it can adjust its position to deposit specimens **accurately** in different locations. Vertical slides allow it to extend and reach high-scoring areas with **ease**. This **adaptability** ensures efficient and consistent performance throughout the match.

Butterfly Drivetrain

Butterfly Drivetrain:

The drivetrain features a **parallel plate** design, with motors placed at the center of the robot to ensure **balanced** weight distribution. These motors are positioned as **low** as possible to the ground, lowering the **center of gravity**, which significantly improves stability and reduces the chance of **tipping** during movement. Moreover, the motors are located **opposite** the vertical slides to create a **counterbalancing** effect. This arrangement helps **maintain** the robot's balance even when the vertical slides are fully **extended**, ensuring that the robot remains steady and **reliable** throughout various tasks on the field.



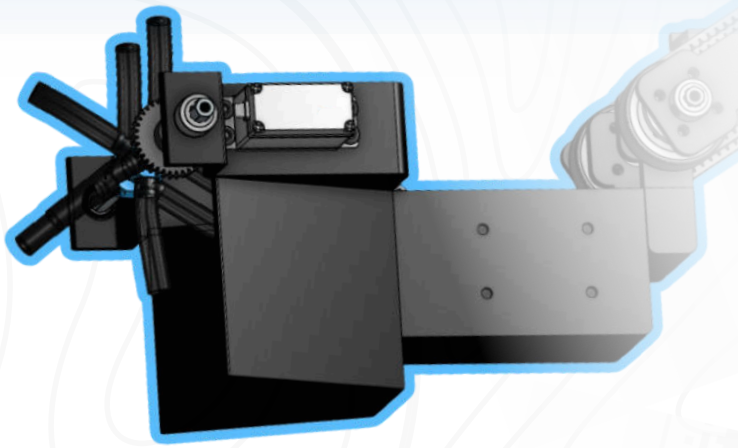
Two Wheel Localizer

Additionally, the drivetrain's **structure** is built for efficiency, offering **smoother** movements and better control during high-speed **maneuvers** or complex tasks. To sum it up, our drivetrain is a grand part of our robot, and could not advance our robot without its design and **functionality**.

Intake Iterations

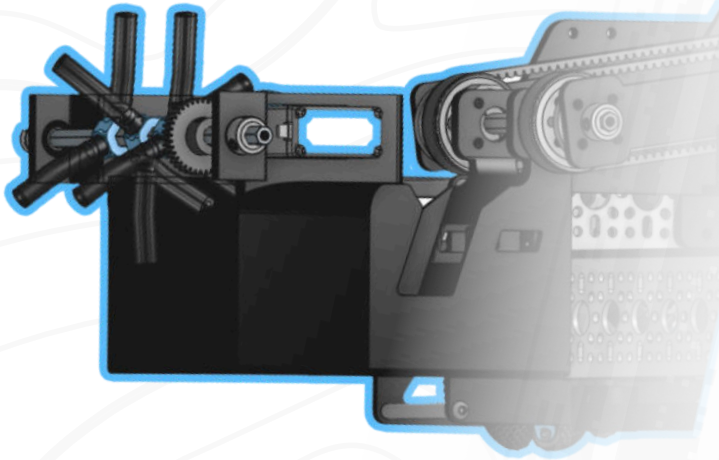
Version 1:

The first version of the **intake** system used a simple intake box but had several flaws. It moved down using a **virtual four-bar** mechanism, keeping the intake's end effector parallel to the ground. However, this caused problems when **pivoting**, making movement inefficient. Samples frequently slipped out, there was no **automated** way to detect colors, and the horizontal sample orientation made it **difficult** for the claw to pick them up **properly**.



Version 2:

Improvements included a front groove to prevent slipping, a color sensor for **detection**, and a vertical sample orientation for easier pickup. However, we still used the **virtual four-bar** mechanism, which continued to cause movement issues, restricting smooth turns. The **servo** intake was still slow and obstructed the claw's path during autonomous **movement**. While better than the first version, this design still struggled with reliable **sample** collection and overall system **automation**.

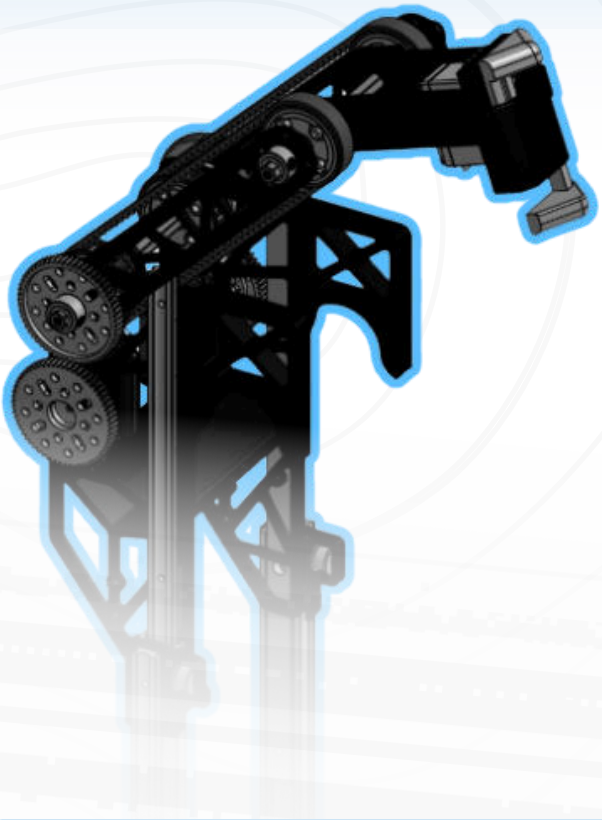


Version 3:

The final design used a **motor-driven** belt with **counter-rotating** rollers and flexible tubing for smooth intake. Instead of the virtual four-bar, we opted to **tilt** the intake down and up at an **angle**, eliminating movement issues. A color sensor identified samples, **closing** a servo gate for correct ones and **ejecting** incorrect ones via additional rollers. The system's tilting mechanism and belt-driven **movement** significantly improved **efficiency**, speed, and automation.



Outtake Iterations



Version 1:

The first outtake design used a **claw** that gripped **samples** from the inside. However, samples often **slipped** out when the robot moved, and the inefficient design **resulted** in fewer game pieces being scored. This version **utilized** a **coaxial virtual four-bar** mechanism for claw movement, providing two degrees of **freedom**: arm rotation and claw rotation. While the claw itself had **issues**, this mechanism was highly efficient for precise **positioning** and remained unchanged in later iterations.

Version 2:

The claw was **redesigned** to grip samples from the outside, improving precision and making it **easier** to secure them. This allowed the robot to move **faster** without the **risk** of dropping samples, leading to better overall **performance**. We **continued** using the coaxial virtual four-bar mechanism from the first version, as it provided smooth and **reliable** movement with **two** degrees of freedom, ensuring optimal arm and claw control **throughout** operation.



Sensors:

The Orca is **equipped** with a variety of sensors that work together to ensure optimal **performance**. One important sensor is the **color** sensor, which helps identify and sort samples after they are **collected**. This sensor ensures that only the **correct** samples, based on the assigned alliance color, are gathered, preventing any **mix-ups** and improving efficiency. Additionally, the Orca uses two **distance** sensors, each capable of measuring up to two **meters**. These sensors assist with maneuvering during the **autonomous** stage by detecting the distance between the two of the **corner** frame walls, **allowing** us the line up perfectly with the high basket. The robot also has a magnet limit switch sensor **attached** to its linkage. This sensor makes sure the linkage stops at the right point, by detecting the **magnetic** field on a magnet attached to the linkage, ensuring they don't **extend** too much or too little, which is **essential** for accurate movement.



Finally, **encoders** on the motors track rotation and speed, providing real-time **data** that helps **control** the robot's movement. This allows for precise, **controlled** motion and smoother operation. Together, these **sensors** enhance the Orca's ability to perform tasks accurately and efficiently, contributing to its overall **functionality**.

Control - Cont.

Code:

The robot's control system is powered by **sophisticated** programming that boosts its efficiency. It processes sensor data in **real-time**, allowing for immediate adjustments during both **autonomous** and teleoperated phases. The **system** features an array of sensors to improve **motor** movement precision, and sensor fusion **enhances** the robot's ability to navigate more effectively. Color **sensors** also support intake and outtake **functions**, ensuring smooth handling of game pieces. The robot **undergoes** continuous testing and refinement to ensure consistent, **reliable** performance during competitions, enabling it to **adapt** to various challenges on the field.

Our program includes different **classes** for controlling motors, sensors, and actuators, as well as for **autonomous** routines. Each class is structured to handle **specific** tasks, such as motor control, sensor **reading**, or navigation logic. This modular **approach** makes the code more organized and easier to modify or **expand**. Key classes interact with each other, ensuring smooth **coordination** and **precise** robot behavior.



Overall Hours:

This season, we have managed to **attain** approximately **210** outreach **hours** through collaborating with our **community**, which includes talking about **STEM** with our school peers and with current **FLL** and potentially future **FTC** members. We were also able to work with **FIRST** to help **host** an event to spread **awareness** about STEM at the Wings Over Camarillo Airshow. We got around **3000** viewers in attendance over the course of **2 days** at this event.

Sister Team Connection:

Starting this season, CSK has had **two** official FTC teams. We have worked together by **sharing** supplies and **learning** from each other in programming, design, and mechanics. Our **collaboration** has **strengthened** both teams and improved our overall performance. We also **combined** outreach efforts to spread awareness about STEM in our community, working **together** to **introduce** more people to robotics and inspire **future** engineers.

Connection with Mentors:

We recognize the value of **mentorship** and professional **connections**. Although we are just beginning this journey, we are seeking **guidance** from engineers, STEM educators, and professionals who can help us improve our **skills** and **knowledge**. Our goal is to build relationships that not only **benefit** our team but also **inspire** others to pursue **careers** in STEM fields. This allows ourselves and our **community** to have a relationship with one-another.

3500 People Reached

7 Members

210 Outreach Hours

∞ Memories









FLL: CSK Robot3rs

The **CSK** foundation not only sponsors **FTC** teams but also two **FLL** (First Lego League) teams. One of the **standout** teams under CSK is the **Robot3rs**, which is mentored by a few members from **our** team. This past season, they recently achieved an impressive victory by **winning** their **regional competition** and earning a spot at the **state championships.** The Robot3rs will soon **aspire** to **join** FTC which will keep the CSK Foundation **involved** with FIRST for a long time.





FLL: CSK Challeng3rs

The second FLL team under the **CSK** Foundation are the CSK **Challeng3rs**, which are also mentored by various members of our **FTC** team. During the 2023-2024 FLL season, they won the **regional competition** for FLL and advanced to the **state championships**. The Challeng3rs have also shown **vast** amounts of **success** on the field. They also won the **Innovation Award in the competition**, which helped secure their place among the ranks of the FLL **community**.

Main Outreach Goal:

During our **many** outreach events, one of our main goals was to **spread** STEM and FIRST everywhere we went. Some ways we did these was by **passing** out **flyers** and **demoing** our robot to show how **fun** STEM can be.



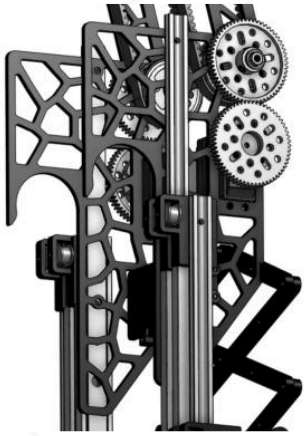
Passing the Word:

Another **strategy** we used to motivate kids to join **robotics** and STEM related activities was through **connections**. Using our parents and teachers, we were able to talk about the **importance** of STEM and how it is useful in the **modern** age.

Additional Projects

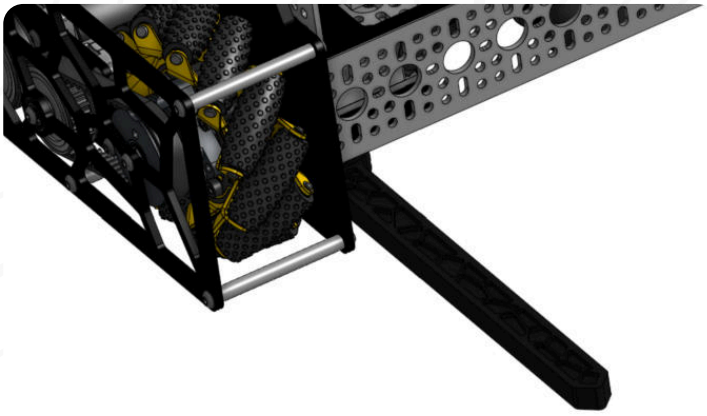
Project 1: Hooks

The Orca utilizes a hook system to **grasp** onto the top bars of the **submersible**, earning additional points during the endgame. Initially, the **hooks** were designed using 3D-printed filament; however, they lacked the **necessary** strength to support the robot's weight. To address this, we **custom**-designed and **manufactured** laser-cut aluminum hooks, significantly improving durability and **performance**.



Project 2: Sweeper

The sweeper **mechanism** optimizes sample collection by **repositioning** pieces for better intake. Without it, the intake would land on **top** of samples, making collection unreliable. To fix this, we designed a 3D-printed sweeper that **rotates** a rod under the **submersible**, allowing the robot to move samples within a 10-inch radius. This prevents **interference** with the intake system, **ensuring** smoother operation.



Project 3: Submersible

Sharing the submersible with our sister team made testing difficult, **limiting** autonomous performance. To address this, we built a wooden **replica** matching the real submersible's **dimensions**. This provided **additional** practice time, allowing us to refine programming and improve **consistency**. The extra testing helped ensure the robot performed reliably in **competition**.



Project 4: Powder Coating

To create a **sleek**, durable design, we opted for laser-cut components rather than traditional metal plates. For further **personalization**, we applied a custom black powder coating by hand, adding both visual **appeal** and strength. This enhanced the robot's durability while maintaining a **professional** look. The powder coating process helped protect components from wear, ensuring **long-lasting** performance.





Season Goals:

Expand Sponsorship Opportunities:
Engage **local** businesses to secure **sponsorships** for team **expenses**. Showcase our **mission** and **achievements** to potential **sponsors** and at outreach events. Use funds to **enhance** our robot, **cover** competition costs, and **support** outreach.

Collaboration with Sister Teams:
Develop a **shared** budget to allocate funds **efficiently**. Share materials to reduce costs and **maximize** sponsorships and **donations**.

Outreach:
Promote FTC through **community** events, workshops, and **demonstrations**. Establish an easy **donation** system and strengthen local **support**. Expand our reach via social **media**, school events, and fundraisers.

Expeditures	2023-24	2024-25
Outreach	$300	$300
Materials	$4,000	$7,000
Other Expenses	$700	$700
Facility Costs	$2,500	$2,500
TOTAL	$7,500	$10,500

Revenue	2023-2024	2024-2025
Sponsorships	0	$500
Donations	$1,000	$7,000
Fundraisers	$200	$1,250
Total	1,200	$8,750

$TR = S + F + D$

S = Sponsorship funds received
F= Fundraising Revenue
D= Donations from supporters

$Cr = M + T + E$

M = Cost of materials and components
T = Cost of tools and equipment
E = Electrical and wiring costs

Finances/Funding Sources:

Throughout the season, our primary goal was to **optimize** spending and **maximize** revenue to sustain our robotics efforts. The lack of sponsors **initially** posed financial challenges, limiting our resources. However, with the support of Softbox LLC, we **secured** funding that **enabled** us to continue **developing** and **enhancing** our robot.

Long Term:

In future seasons, we aim to **further** strengthen our financial strategy by **increasing** revenue and **minimizing** expenses. One key approach is securing additional sponsorships, which would reduce our **reliance** on team member entrance fees as a **primary** funding source. This would make participation more **accessible** while ensuring sustainable **growth** through sponsorships and fundraising efforts.